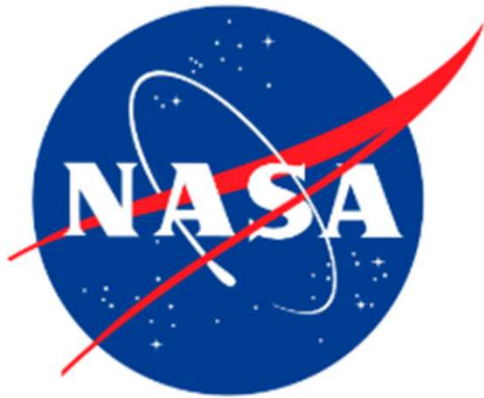




WHEN TRUST MATTERS

The NASA and DNV Challenge on Optimization under Uncertainty: Part II

- The physical system behind the use-case
- Scoring of the submitted solutions



Use-Case: Wind Farm Optimization

Goal: Maximize power production

Constraint: Minimize maintenance cost and risks of turbine breakdown

Challenge: "Optimization under uncertainty"

- Yaw adjustment and optimal turbine placement reduces wake effects → increased power.
- But: Increases mechanical strain on the turbine → higher wear and a greater risk of failure.
- Data is limited, and simulations are costly and time consuming

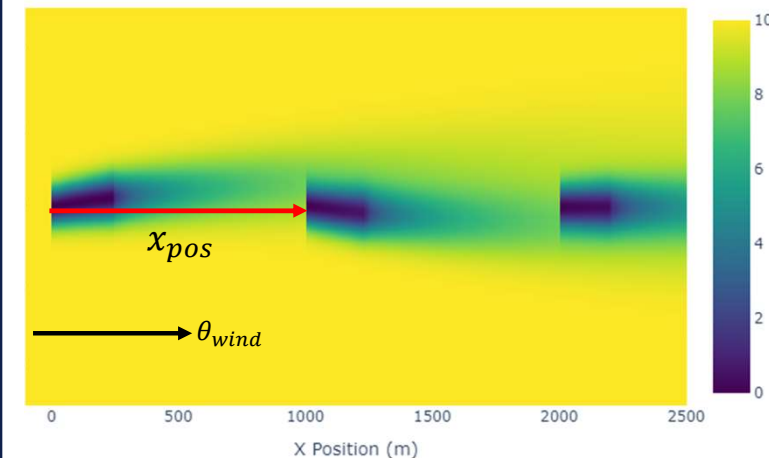
Solution: Optimization + Uncertainty Quantification

Why this use-case?

- DNV: Understand wake effects to improve power output and reduce wear/risk
- NASA: Wake effects critical in flight formation for fuel efficiency and minimizing turbulence
- Inverse problem insights: Reduce wake effects in wind farms $\leftrightarrow$ Exploit wake effects in flight optimization

Control parameters, x_c :

Yaw angle and spacing between turbines



Uncertainty sources, x_a and x_e

Aleatory: Wind speed, variability and direction

Epistemic: Physical/atmospheric constants determining wake deflection and decay

Scoring Procedure

Each team is evaluated across four of the subtasks:

1. Estimating the distribution of aleatory variables, $X_a \rightarrow$ KL-divergence to true distribution
2. Estimating the epistemic variables, $X_e : \rightarrow$ Euclidean distance to true X_e
3. Performance based design: $\rightarrow$ Power Output
4. Reliability based design: $\rightarrow$ Probability of Failure (note: we apply a log-transform before scoring:
 $x_{POF} \rightarrow \log 10(x_{POF}))$

To combine scores from all tasks, we normalize all metrics to a scale from 0-100 points

$$\text{score} = 100 \times \frac{\delta(x - x_{\min}) + (1 - \delta)(x_{\max} - x)}{x_{\max} - x_{\min}}$$

where x is the raw value of the metric for each subtask, $x_{\min/\max}$ is the min/max value across all teams, and $\delta \in [0,1]$ if lower/higher metric is better, respectively.

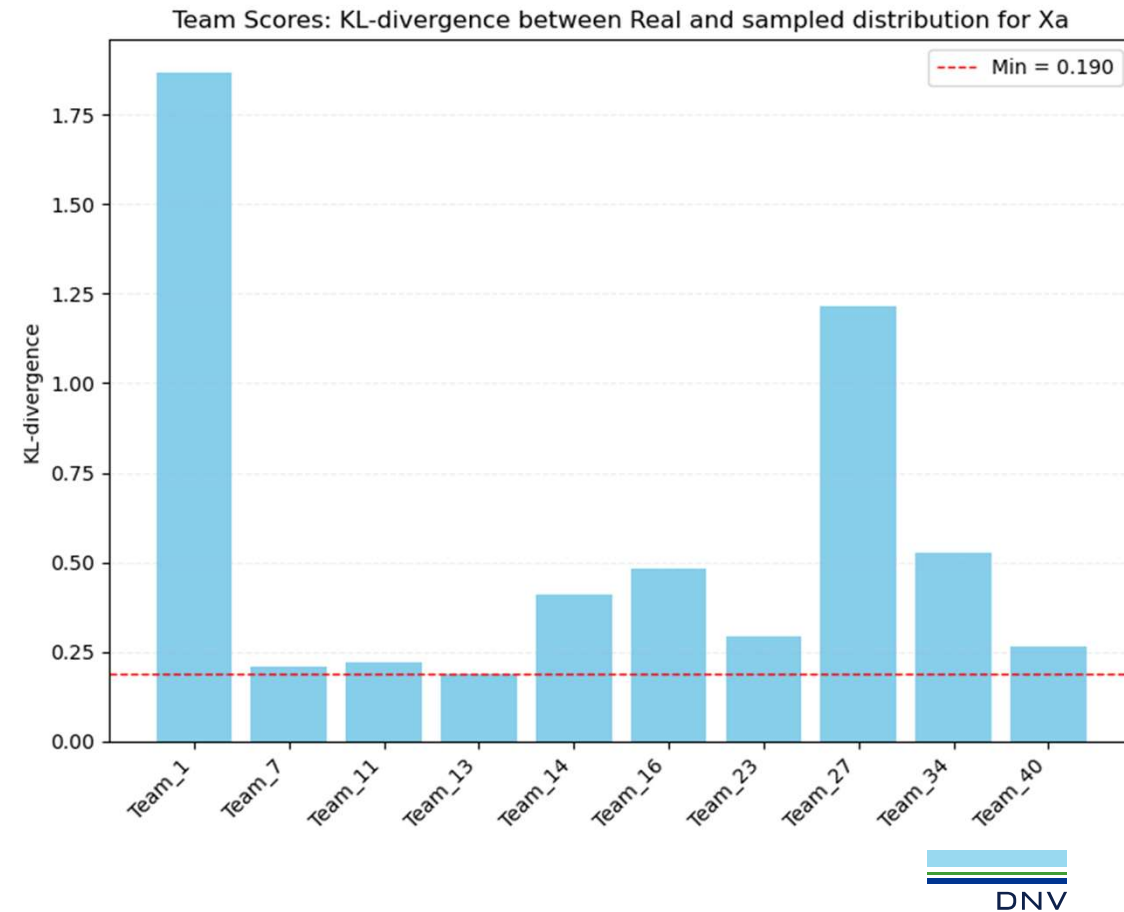
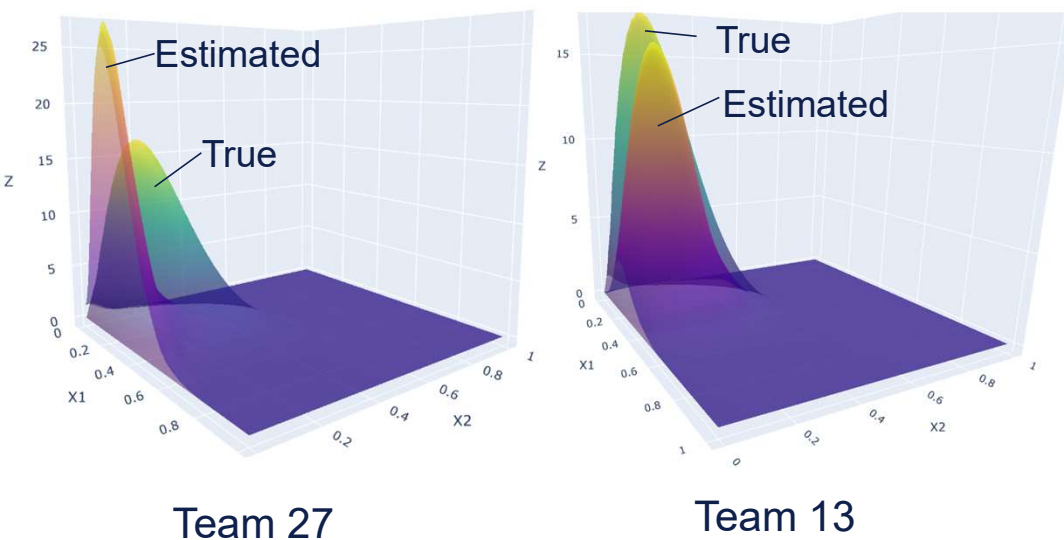
This means:

- The best team on a given subtask gets 100 points
- The worst team gets 0 points
- Others are scaled linearly in between

Scoring for Estimated X_a

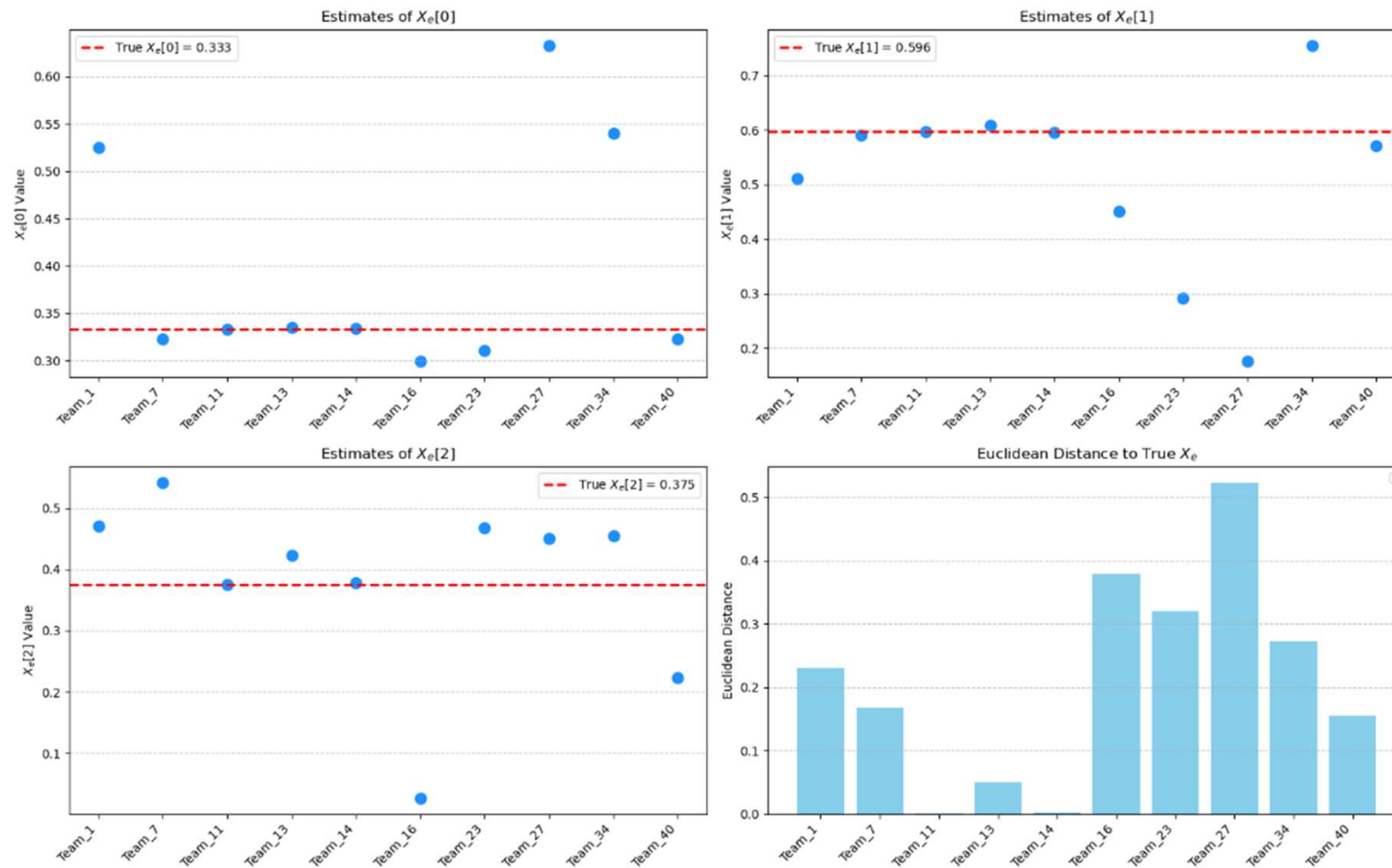
We calculate the KL-divergence between the “true” and estimated distribution of X_a , (based on the $N = 10^4$ submitted samples from each team)

Example: “true” vs estimated pdf for X_a



Scoring for Estimated X_e

We calculate the Euclidean distance between the “best guess estimates” and the true components of X_e



Combined Scoring for Uncertainty Model



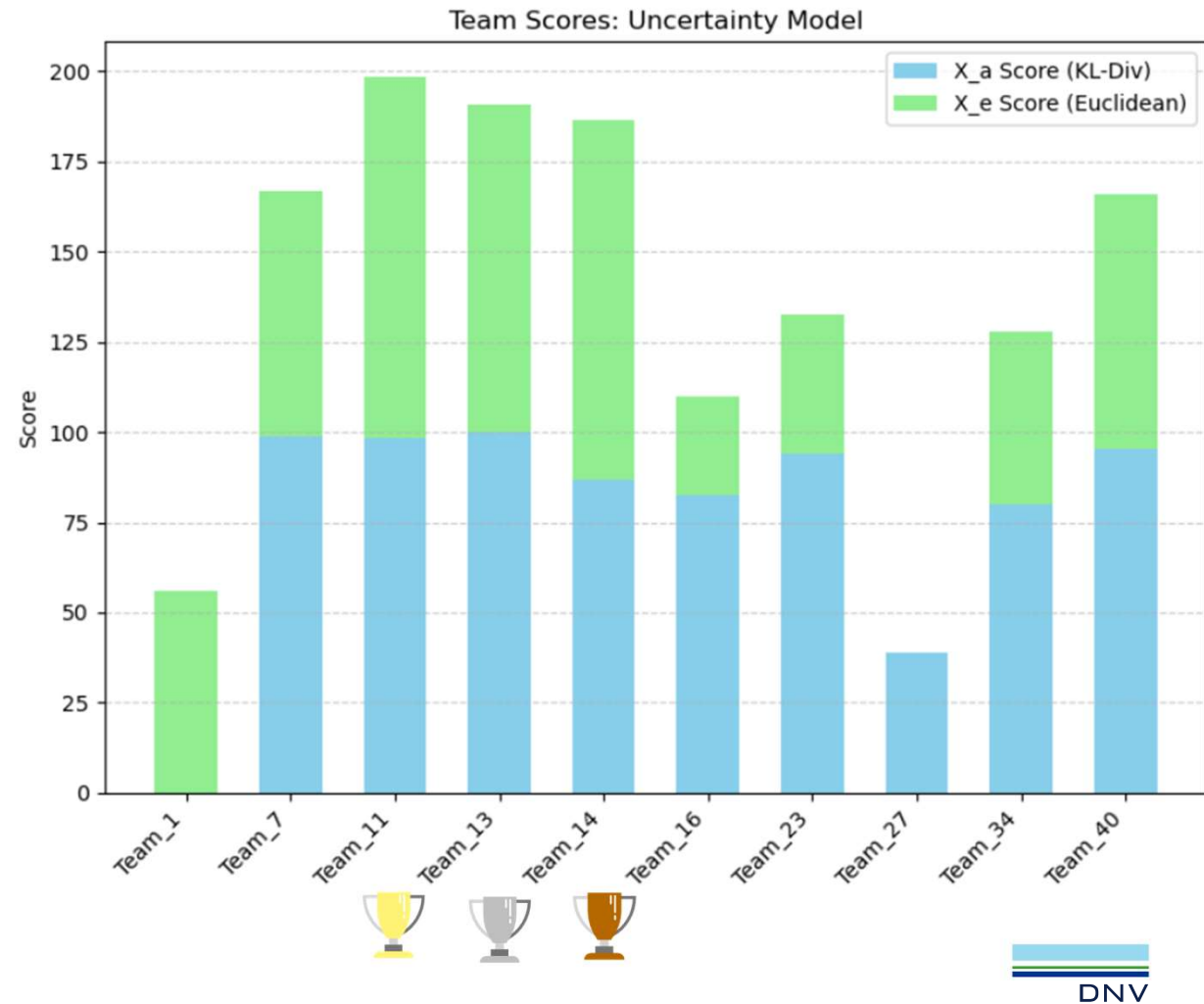
Team 11: Julien Demange-Chryst et al.



Team 13: Matteo Broggi et al.



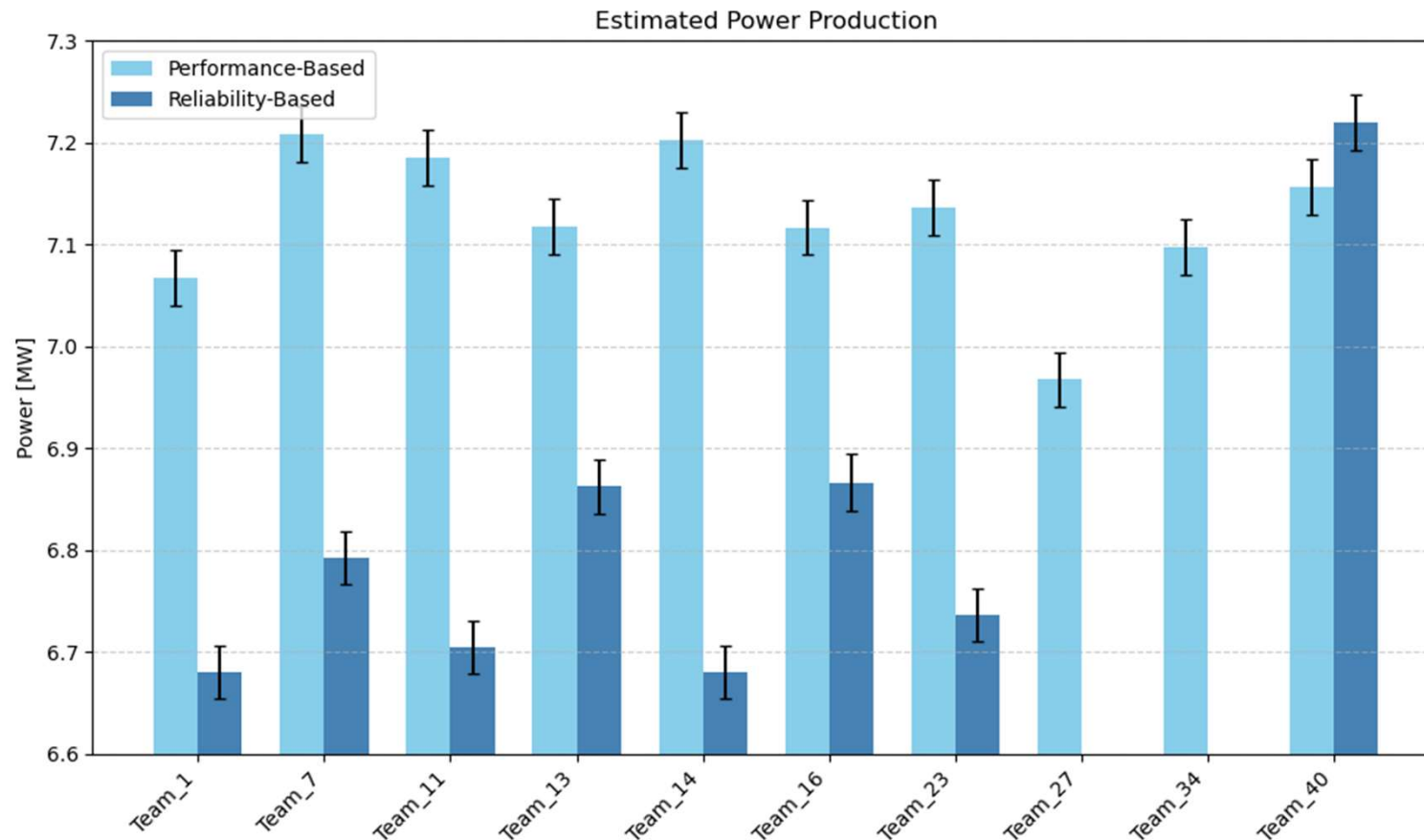
Team 14: Roberto Rocchetta et al.



Scoring of Performance-Based Design

We calculate the estimated Power Production, based on the suggested control parameters, X_c , from each team.

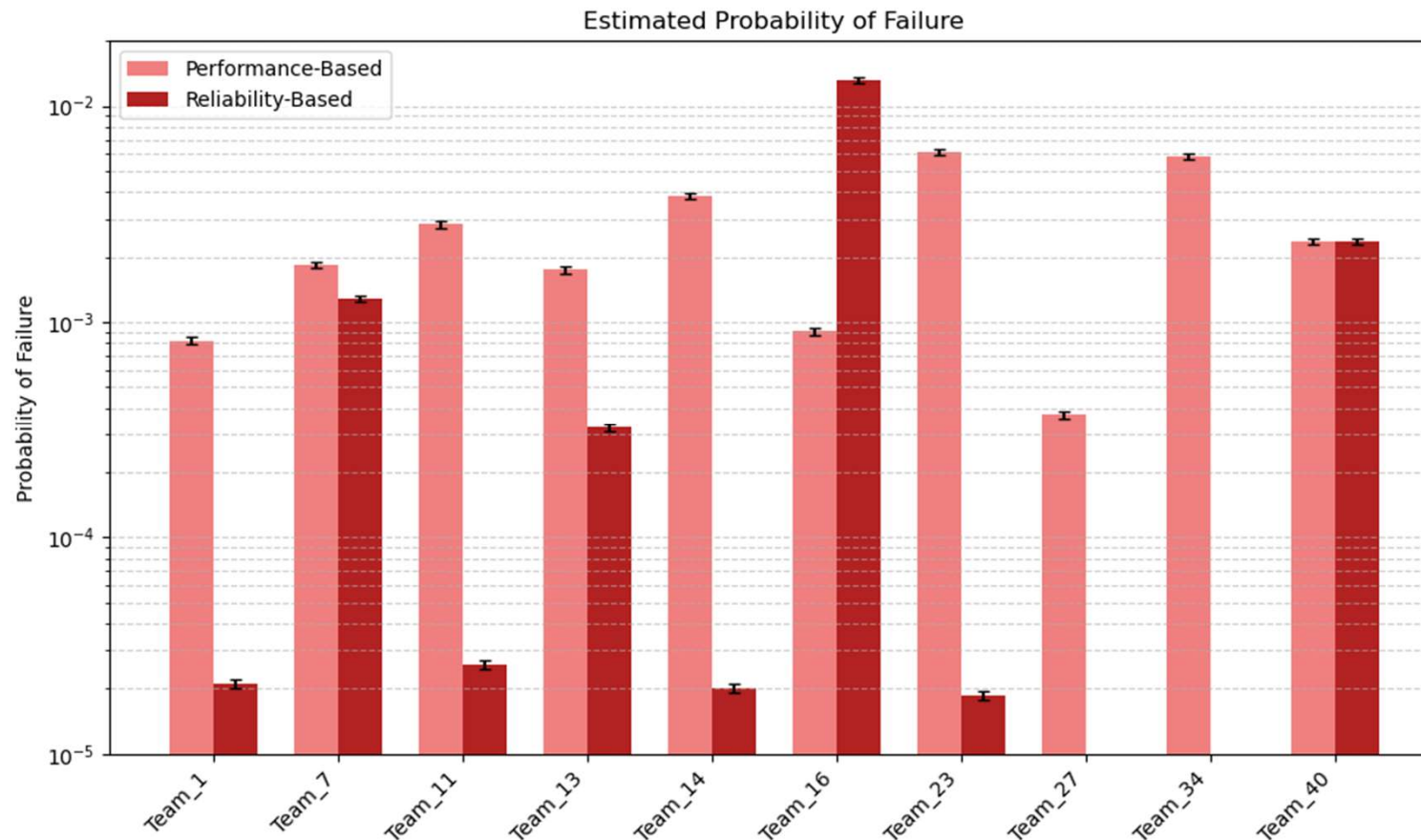
The proposed designs are evaluated on the “true system”, with the correct values for X_e and distribution for X_a



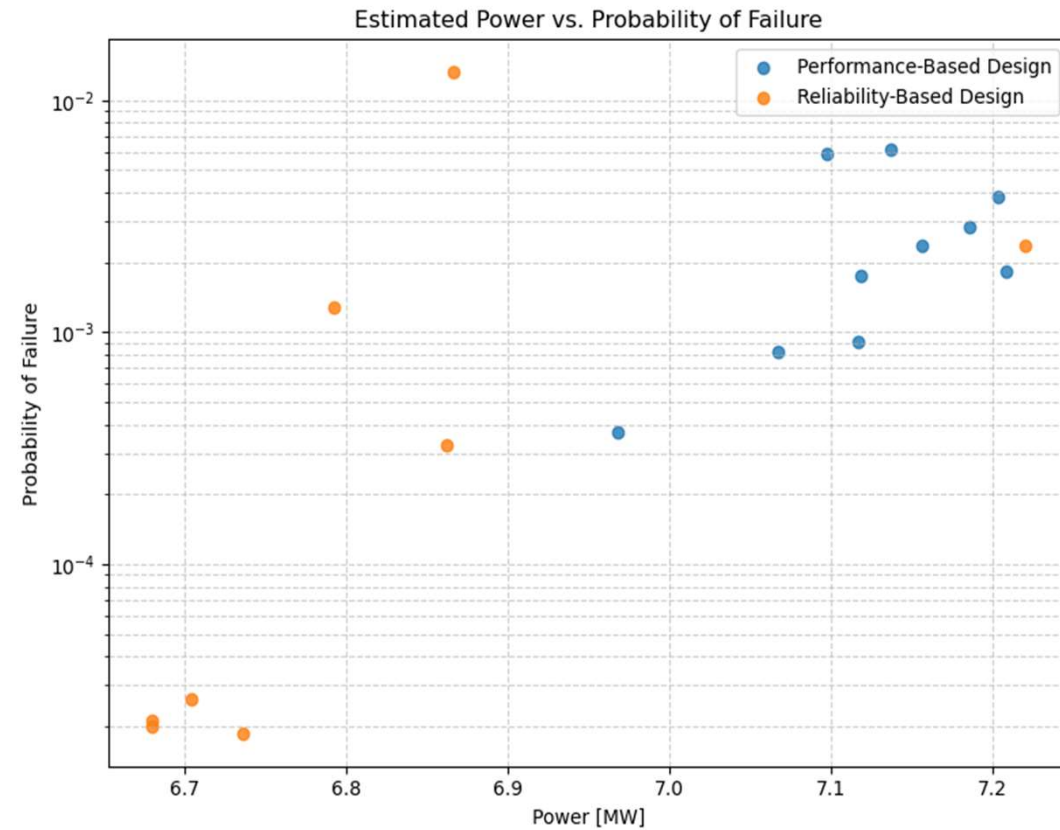
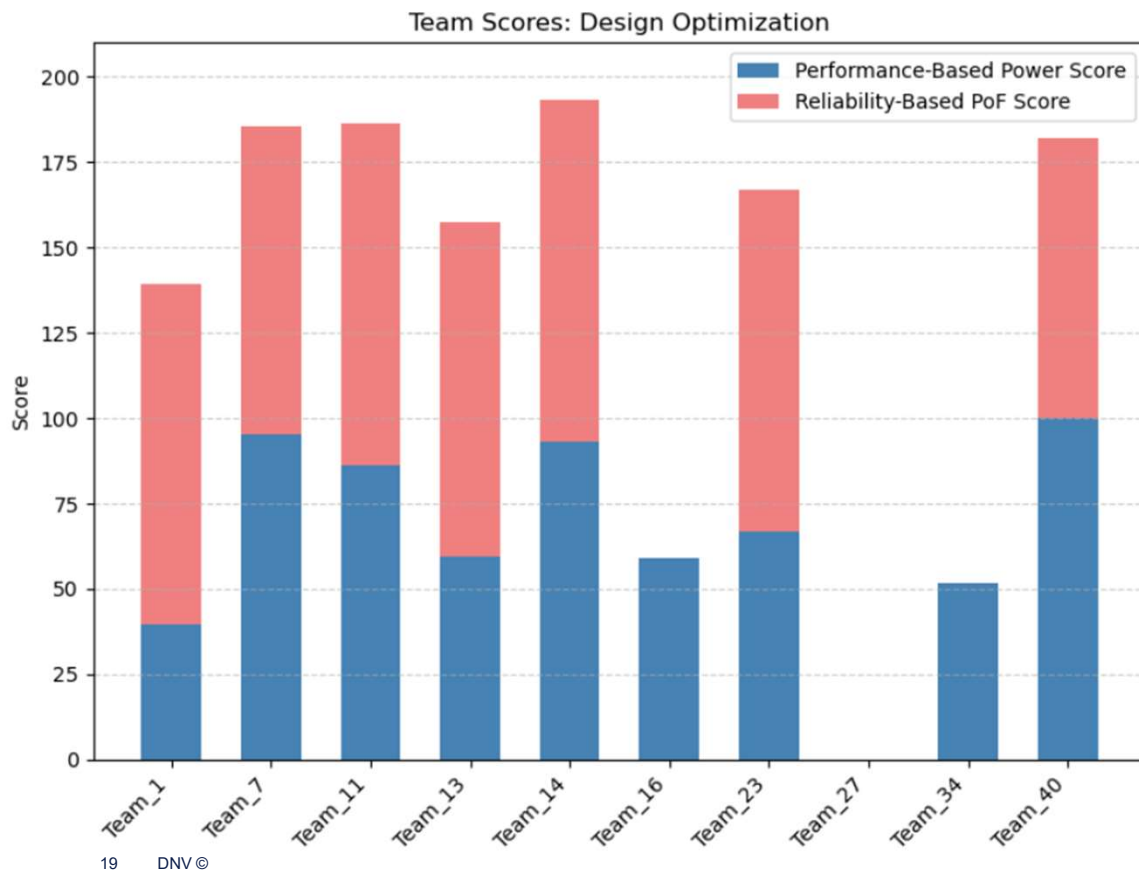
Scoring of Reliability-Based Design

We calculate the estimated probability of failure based on the suggested control parameters, X_c , from each team.

The proposed designs are evaluated on the “true system”, with the correct values for X_e and distribution for X_a



Combined Scoring for Design Optimization



Combined Scoring for Design Optimization



Team 14: Roberto Rocchetta et al.

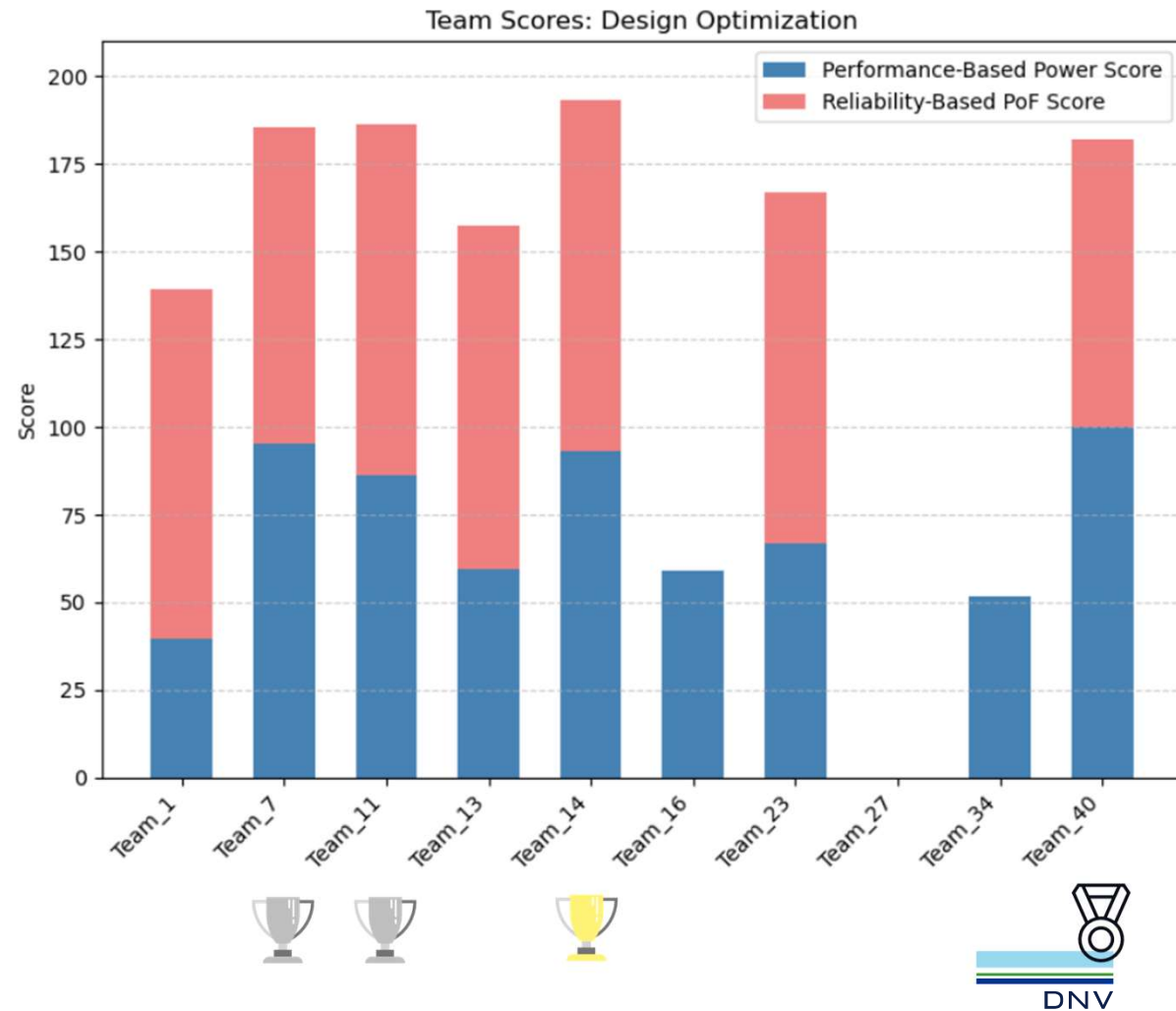


Team 11: Julien Demange-Chryst et al.

Team 7: Somdatta Goswami et al.



Team 40: Tairan Wang et al.



Combined Score Across the Four Subtasks:



Team 11: Julien Demange-Chryst et al.

Team 14: Roberto Rocchetta et al.



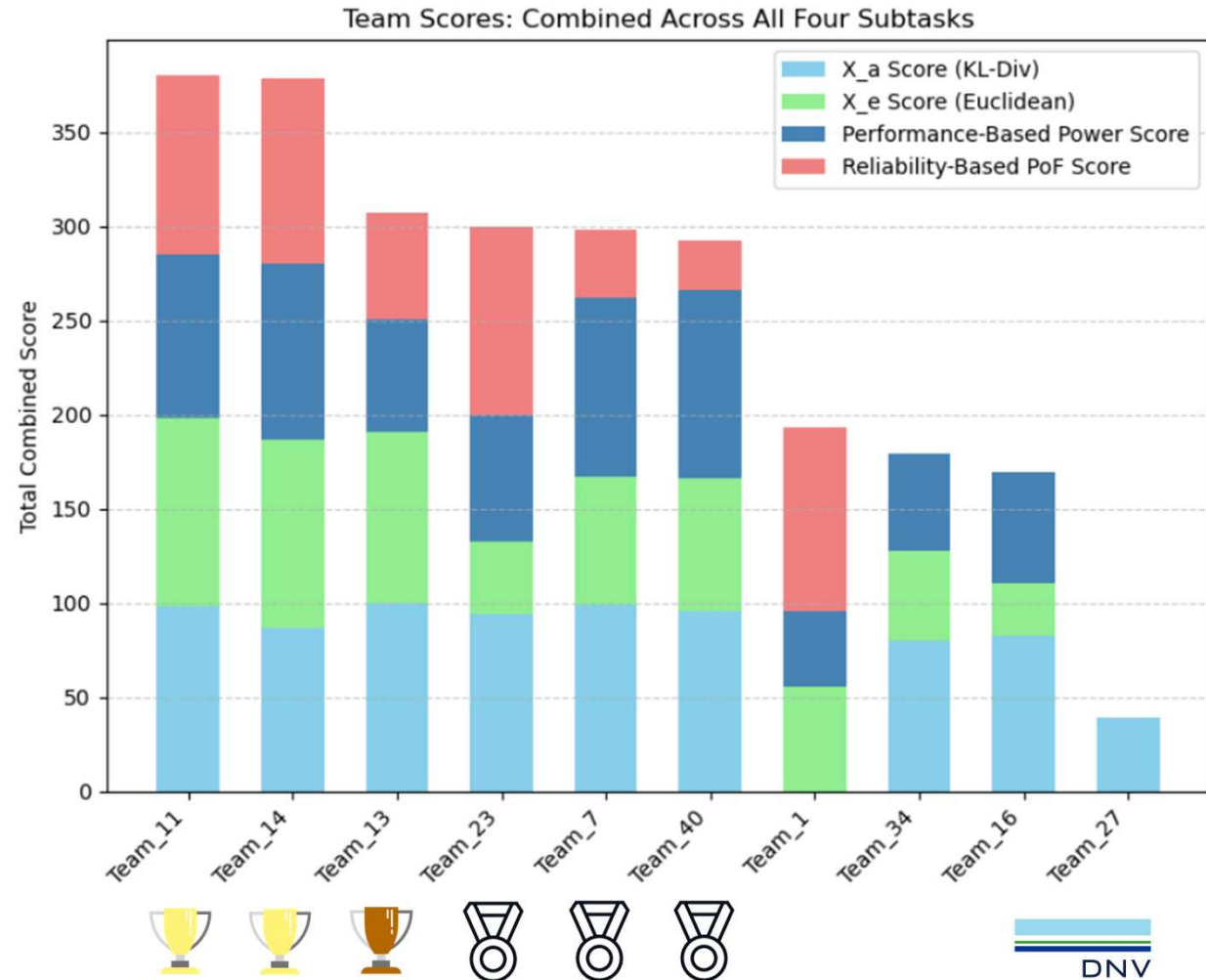
Team 13: Matteo Broggi et al.



Team 23: Masaru Kitahara et al.

Team 7: Somdatta Goswami et al.

Team 40: Tairan Wang et al.



The NASA and DNV Challenge on Optimization under Uncertainty

NASA and DNV launched a joint challenge to the community to improve how we design and control complex systems—like spacecraft or offshore structures—under uncertainty.

The challenge invited participants from across many different scientific and engineering fields

Challenge Outcome:

- Wide range of novel methods to address this problem
- Top 10 contributions submitted detailed papers describing their approach.
- Establishing networks with top research communities

Thanks to all the teams for the participation!

